Engineering Materials MS4092

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Number of problems: 8
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Duration: 24 Minutes

What is the term for the ratio of stress to strain below the yield strength?

- ○ Tensile strength
- ◉ Young's modulus
- ○ Poisson's ratio
- ○ Hooke's constant

Clear selection

The crystal structure of hard Martensite phase in steel is

- ◉ BCT body-centered tetragonal
- ○ BCC body-centered cubic
- ○ HCP hexagonal close packed
- ○ FCC face-centered cubic

Clear selection

Choose 2 of the following materials which not shown ductile-to-brittle transition if used in cryogenic temperature

- ( ) Austenitic stainless steel ✓
- (•) Aluminium
- ( ) Martensitic Steel
- ( ) High-chromium alloyed steel
- ( ) Malleable cast iron

Clear selection

What is the consequence of the increasing of dislocation density in metals after tensile test?

- ( ) Materials become softer
- ( ) No difference
- ( ) Elasticity increased
- (•) Hardness increased

Clear selection

Steel is more ductile than Tungsten Carbide. Ductility can be recognise from:

- ( ) Modulus of Elasticity of Tungsten Carbide is lower than of steel.
- ( ) Stress vs elongation curve. Elongation of WC is bigger than steel.
- ( ) Impact value vs Temperature curve. Impact value of WC is higher than steel.
- (•) Stress vs elongation curve. Elongation of steel is bigger than WC.

Clear selection

stress ↑ ⇒ makin brittle

elongation ↑ ⇒ makin ductile

The following items is the hardest phase in carbon steel microstructure

- ( ) Ferrite
- ( ) Cementite
- ( ) Pearlite
- (•) Martensite
- ( ) Austenite

martensite > ferrite > pearlite > austenite
||
cementite → lebih malleable

Clear selection

A cast iron is relatively easier and cheaper to manufacture for small industries than steel is due to:

- (•) Lower melting temperature which is 1143 C
- ( ) Lower melting temperature which is 700 C
- ( ) Lower carbon content
- ( ) High melting temperature which is 1535 C

melting temp cast iron sekitar 1150°C

Clear selection

What is the main chemical composition for austenitic stainless steel?

- ( ) Fe-C-Mn
- ( ) Fe3C-C-Si
- ( ) Fe-Co-Ni
- ( ) Fe-C
- (•) Fe-Cr-Mn → ini khas stainless steel
- ( ) Fe-Cr-Ni

Clear selection

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